GLA UNIVERSITY

Department of Computer Engineering & Applications

MINI PROJECT – 1 SYNOPSIS

1. BASIC PROJECT DETAILS

Project Title	Capstone: Unified Product Experience Ecosystem
Batch No.	Batch 6
Section	3AG
Semester	5th Sem
Date of Submission	18/09/2026
Faculty Mentor / Guide	Mr. Satyam Kumar Jha

2. TEAM MEMBERS & ROLES

List every team member, their university roll number, and their specific role/responsibility in the project (e.g., Frontend Development, UI/UX Design, Backend & API, Testing & Documentation, Research & Presentation).

S.No.	Name of Student	University Roll No.	Role in Project
1	Krishna Goyal	2415000837	Team Lead & Catalog Module Development (Frontend)
2	Kunal Goyal	2415000867	User Dashboard Development & UI/UX Design
3	Krati Verma	2415000828	Admin Console Development (Frontend)
4	Kuldeep Pal	2415000862	Routing, Navigation & Cross-Section Integration
5	Krishna Kumar Tiwari	2415000843	Design System, Theming & Testing/Documentation

3. PROJECT DESCRIPTION

Briefly describe what the project is about — the problem it addresses and the core idea behind it (150–200 words).

Modern digital products rarely exist as a single page or flow — they are made up of multiple experiences, such as a customer-facing catalog, a personalized user dashboard, and an internal admin console, that must still work together as one cohesive product. Building these as isolated modules often leads to duplicated logic, inconsistent design, and data that falls out of sync, especially when no backend is available to coordinate them.

This project, *Unified Product Experience Ecosystem*, addresses this by building a purely frontend ecosystem that unifies catalog browsing, a user dashboard, and an admin console under a single design system. All three sections are independently routed, yet share a common in-browser data layer, ensuring that actions in one section (such as an admin updating a product) are reflected consistently across the others.

The core idea is to demonstrate advanced client-side state management, consistent theming, and smooth transitions between sections — all without relying on a server. By simulating real-world multi-role product architecture entirely on the frontend, the project highlights how thoughtful architecture and design discipline can deliver a seamless, unified user experience using client-side technologies alone.

4. OBJECTIVES & PROPOSED WORK (WHAT WE ARE GOING TO DO)

Explain the key features you plan to build, the technology/approach you will use, and the current stage of work. Use bullet points if needed.

- Build a **Catalog Module** with product listing, search, filtering, and detailed product view pages.
- Build a **User Dashboard** showing saved/favorited products, simulated order history, and editable profile settings.
- Build an **Admin Console** to add, edit, or delete products and view simulated analytics.
- Develop a **shared in-browser data layer** so that changes made in one section (e.g., Admin) instantly reflect across the others (e.g., Catalog, Dashboard).
- Design a **unified design system** with reusable components (buttons, cards, modals, navbars) and consistent theming (light/dark mode) across all sections.
- Implement **independent client-side routing** for each section with smooth transitions between them.
- **Approach:** React.js for the frontend, Context API/Redux for state management, React Router for navigation, Tailwind CSS for styling, and localStorage/IndexedDB for data persistence — no backend server involved.
- **Current Stage:** Problem statement, architecture, and tech stack finalized; feature planning in progress; development yet to begin.

5. FUTURE SCOPE / ENHANCEMENTS

Describe what you plan to add or improve in the future beyond the current scope of the project.

- Integrate a real backend (Node.js/Express with a database) to replace in-browser storage for true persistence and multi-device sync.
- Add user authentication and role-based access control (customer vs admin login).
- Introduce real-time notifications (e.g., order updates, stock alerts) using WebSockets
- Add advanced analytics and data visualization in the Admin Console (charts, sales trends).
- Make the ecosystem a Progressive Web App (PWA) for offline access and mobile installability.
- Support multi-language and accessibility (a11y) improvements for wider usability.

6. TOOLS & TECHNOLOGIES USED

- **Frontend:** React.js, React Router, Tailwind CSS
- **Development Tools:** Vite, Git & GitHub, VS Code

7. EXPECTED OUTCOME

- A fully functional, purely frontend web application integrating Catalog, Dashboard, and Admin sections into one cohesive product ecosystem.
- Seamless navigation and smooth transitions between independently routed sections.
- Consistent design language and theming maintained throughout the application.
- Synchronized, shared state across all sections without any backend/server dependency.

Signature of Faculty Mentor